OFFICIAL PUBLIC SYSTEM GUIDE · 23 OCTOBER 2026

DataChips, Circuits & Circuit Genomes

How verified reusable computational capabilities become larger systems

BTG-LIB-CIRCUITS-SG · Edition 1.0 · Audience: System owners, researchers, institutional evaluators and authorized nontechnical operators · Issued 23 October 2026

Publication boundary: This is an explanation of the system and its public operating model, not a source release, engineering specification, certification or promise of live service availability. Product status is stated with the limitations below.

Executive overview

A DataChip is a reusable computational capability with a stated purpose and boundaries. A Circuit connects compatible capabilities to complete a larger task. A Circuit Genome records which components, versions and approved relationships belong to that assembly. This handbook concerns use, accountability and qualification—not program source or implementation APIs.

01 — Components, circuits and genomes

A DataChip may perform a calculation, classification, measurement, reconstruction or bounded inference. Each one needs an identity, supported input class, expected output, limitations and an evidence record.

A Circuit is the working arrangement of multiple chips. A Genome preserves the versioned composition and the rules required to identify or restore it. A diagram is a way to understand the system; it is not by itself proof of correct execution.

02 — Where components come from

A qualified capability can originate in conventional code, an approved scientific process, an existing BTG subsystem or a candidate proposed by NIKI. Provenance and license constraints travel with its record.

A model-generated suggestion, copied fragment or visually assembled card must not be treated as a novel, owned invention until authorship, rights, originality and test evidence have been established.

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03 — What a trustworthy chip record contains

A reader should expect a plain-language purpose, version, approved use, unit or data assumptions, accepted and rejected input categories, required dependencies, error behavior, quality evidence and owner or custodian.

Evaluation must specify whether a result is known correct, statistically supported, limited to a bounded test range or not verified. High test counts are meaningful only within the stated dataset and oracle.

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04 — Connecting chips safely

Two components may be individually valid yet incompatible because their units, input shapes, timing, data permissions or error handling differ. The combined Circuit must therefore be tested as a system.

Responsible assembly considers termination, resource use, recoverability, incorrect output propagation and whether a step could access material outside its permitted mandate.

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05 — Circuit Genomes and versions

A Genome is a traceable recipe for a particular configuration of components and their connections. It supports evaluation of what changed, what remained qualified and whether a previously tested configuration can be reconstructed.

Replacing one chip, dependency or data source can change system behavior. An older qualification should not be silently carried forward to a materially different Genome.

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06 — The Composer experience

BTG’s Composer is intended to present conventional source editing and visual composition together. Users may inspect a graph, understand dependencies and assemble permitted components under controlled conditions.

A drag-and-drop composition is not the same as an automatically verified program. The operator must see unconnected inputs, missing components, unsupported operations and unresolved rights before approval.

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07 — Human authority and automated invention

NIKI may propose a new chip or repair. It should preserve candidate evidence and failures, then submit outputs to independent tests. ADAM and ENTITY govern whether a result can be promoted or used for a protected purpose.

Automatically generated algorithms do not receive ownership, safety certification or unrestricted deployment authority merely because they compile or pass internal tests.

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08 — Operations and incident response

Operational records should identify a Circuit version, operator mandate, material input category, run outcome, resource limits and relevant error or integrity evidence. Users should be able to isolate a failed chip rather than losing the provenance of an entire run.

Where private source material or personal data is involved, incident reports must expose only what authorized reviewers need; public documentation should contain no implementation secrets.

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09 — Commercial rights and valuation

A reproducible Circuit could have value as a licensed capability, transferred intellectual property or part of a permitted investment structure. Those are not identical transactions.

Commercial admission requires verifiable rights, useful performance, reproducibility, contract restrictions and truthful disclosure of unproven features. A visual graph alone is not a market asset.

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10 — Qualification and current limitations

BTG publicly identifies Composer and runtime engineering demonstrations with bounded source-parity and test scopes. Those records do not establish universal conversion of arbitrary programs or unrestricted production safety.

Before operational use, the specific version must have its own release, compatibility and security evidence. The complete proprietary implementation is not published in this library.

Related manuals and use

The companion User & Operator Manual describes the decisions and reviews applicable to this system. References to proposed user flows are explicitly conditional on qualification of the relevant product release.

Updated 23 October 2026 · Editorial documentation, not proof of regulatory authorization or release readiness · Proprietary implementation withheld

Expanded system explanation

Question	Clear answer
Audience and purpose	DataChips, Circuits & Circuit Genomes is described for the public, institutional evaluators and prospective partners; this is not a software implementation reference.
What exists	A component system where tested capabilities form compatible executable workflows, with records of versions, owners, test results and permitted use.
Analogy	A DataChip is a tested tool; a Circuit is a connected assembly of tools; a Genome is the versioned recipe identifying which tools and connections make the assembly.
Authority	The described process requires actual contracts, human authority and technical qualification at every relevant boundary.

Practical examples — what a reader should infer

Simple circuit

A temperature-reading chip sends a numerical measurement to a threshold chip, then to a light-indicator chip. Testing each alone does not prove the assembled alert always works; units, delays and missing readings also matter.

Genome revision

Replacing a sensor with one reporting Fahrenheit instead of Celsius is a meaningful change. The new Genome needs explicit conversion, acceptance tests and a new evidence reference.

Economic example

A validated reusable industrial routine may support a limited execution licence; commercial value must be supported by lawful title, demand and evidence, not merely the name “DataChip”.

System lifecycle — each separate decision

1. Describe an individual chip’s intended inputs and outputs
2. Identify the authorized origin, version and dependencies
3. Verify the chip under stated test cases
4. Connect compatible chips under a defined Circuit
5. Test end-to-end behaviour, failures and permissions
6. Freeze the qualified Circuit Genome and record change/retest decisions

People, organizations and responsible roles

- Component owner defines scope and rights
- Composer operator proposes a visual or text-based composition
- Reviewer examines contracts, compatibility and test evidence
- Governance authority decides whether a version may be retained or deployed

Where the system can fail

- Incompatible types or units can silently corrupt a result.
- A passing isolated component may fail in an assembly.
- Model-produced code and imported source can contain security or licensing risks.
- Composer conversion claims require source parity and test qualification; arbitrary software conversion is not yet proven.

Evidence a serious evaluator would require

These are acceptance requirements, not claims that each artifact has been issued or independently certified:

1. Signed or recorded owner/lineage statement
2. Versioned input-output contract and supported environment
3. Unit/integration/regression tests including negative cases
4. Qualification and admitted-runtime boundaries
5. Reproducible build identity without revealing private source
6. Rights Passport and commercial contract when applicable

Terms explained in ordinary language

Term	Meaning
DataChip	A defined reusable computation with declared behavioural boundaries.
Circuit	A set of connected components executing as one workflow.

Term	Meaning
Circuit Genome	A specific, versioned assembly description.
Parity	Observed equivalence against the required original behaviour.

How to read the current status accurately

The architecture and local engineering records are distinct from production operation. A named component or test figure cannot establish authorized commercial rights, fully qualified infrastructure or independently accepted financial activity. Read the dated corporate evidence register for each claimed version; ask for a scoped, access-controlled evidence bundle for decisions that rely on private implementation.

More detailed reading: User & Operator Manual · Non-source engineering evaluation · Regulatory assessment pathway · Complete cross-system example.